

Kidney paired donation and optimizing the use of live donor organs.

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Evidence abstract

CONTEXT Blood type and crossmatch incompatibility will exclude at least one third of patients in need from receiving a live donor kidney transplant.

Kidney paired donation (KPD) offers incompatible donor/recipient pairs the opportunity to match for compatible transplants.

Despite its increasing popularity, very few transplants have resulted from KPD.

OBJECTIVE To determine the potential impact of improved matching schemes on the number and quality of transplants achievable with KPD.

DESIGN, SETTING, AND POPULATION We developed a model that simulates pools of incompatible donor/recipient pairs.

We designed a mathematically verifiable optimized matching algorithm and compared it with the scheme currently used in some centers and regions.

Simulated patients from the general community with characteristics drawn from distributions describing end-stage renal disease patients eligible for renal transplantation and their willing and eligible live donors.

MAIN OUTCOME MEASURES Number of kidneys matched, HLA mismatch of matched kidneys, and number of grafts surviving 5 years after transplantation.

RESULTS A national optimized matching algorithm would result in more transplants (47.7% vs 42.0%, $P < .001$), better HLA concordance (3.0 vs 4.5 mismatched antigens; $P < .001$), more grafts surviving at 5 years (34.9% vs 28.7%; $P < .001$), and a reduction in the number of pairs required to travel (2.9% vs 18.4%; $P < .001$) when compared with an extension of the currently used first-accept scheme to a national level.

Furthermore, highly sensitized patients would benefit 6-fold from a national optimized scheme (2.3% vs 14.1% successfully matched; $P < .001$).

Even if only 7% of patients awaiting kidney transplantation participated in an optimized national KPD program, the health care system could save as much as \$750 million.

CONCLUSIONS The combination of a national KPD program and a mathematically optimized matching algorithm yields more matches with lower HLA disparity.

Optimized matching affords patients the flexibility of customizing their matching priorities and the security of knowing that the greatest number of high-quality matches will be found and distributed equitably.

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